AIDCF-018-PHY Physical Site, Building and Security Infrastructure Domain Guide

Document control

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Revision history

Version	Date	Summary
1.0	2026-10-10	Initial issue: domain guide for the 24 PHY controls, gate linkage, parameters, evidence and testing-safety expectations.

1 Purpose and scope

This guide explains how the PHY domain of AIDCF-007A is applied when assessing whether an AI Capability Instance can achieve its declared Service Outcomes. It covers the site, the building and its structure, physical zoning, access and surveillance, custody of assets and media, and fire, life-safety and emergency interfaces.

The PHY domain answers four questions for the assurance case:

1. Is the site exposed to natural or man-made events that could disable redundant infrastructure together?
2. Can the building physically carry, house and separate the dense, liquid-cooled and battery-laden equipment of AI capability?
3. Are people, plant and AI assets protected against unauthorised access, theft and tampering?
4. Will fire be detected and controlled, and people escape, without the emergency response itself causing an avoidable loss of capability?

In scope: site hazard assessment and external exposures; logistics routes; adjacent occupancies; structural capacity, rack loading, anchorage, vibration and envelope; security, fire, water and tenant zoning; physical access

control, visitor management, video surveillance and alarm response; delivery, movement, media custody and decommissioning; fire detection, suppression interfaces, egress and emergency shutdown.

Out of scope, with interfaces: digital access to physical-security systems (SEC), electrical arc-flash and battery-system hazards (POW), refrigerants and battery materials handling (ENV), leak detection and liquid isolation functions (THM), hardware provenance (SUP).

This guide is informative except where a requirement carries a DG-PHY-REQ identifier.

2 Why this domain matters for AI capability

Physical design assumptions that were adequate for enterprise data centres are being exceeded by AI deployments in ways that are easy to miss.

Mass and wet mass. Rack-scale accelerator systems with integrated liquid-cooling manifolds weigh far more than legacy racks, and the operating mass includes coolant in cold plates, manifolds, rack-level CDUs and the secondary pipework above or below the rack. Immersion tanks concentrate several tonnes on a small footprint. Structural capacity statements prepared for air-cooled halls frequently use dry shipping mass or uniformly distributed loads that do not represent concentrated rack positions (MCF-PHY-005 (Structural Capacity), MCF-PHY-006 (Rack Loading)).

Logistics. A fully populated rack shipped as one integrated unit must travel from dock to position through doors, lifts, ramps and corridors rated for it. A site that cannot receive the rack without dismantling it has a capability constraint that may only be discovered at delivery (MCF-PHY-003 (Access and Logistics)).

Liquid in the white space. Liquid cooling brings water, glycol or dielectric fluid into rooms that also contain busway and PDUs at high current. Water zoning — the spatial and physical separation of liquid-bearing systems from electrical systems — becomes a first-order design control (MCF-PHY-011 (Water Zones)).

Lithium-ion everywhere. Lithium-ion UPS strings, BESS and rack battery backup units place stored chemical energy throughout the facility. Detection must find off-gassing before thermal runaway, compartments must contain a battery fire, and responders must know where the batteries are (MCF-PHY-010 (Fire Zones), MCF-PHY-021 (Fire Detection), MCF-PHY-023 (Egress and Emergency Access)).

Fire response versus cooling continuity. Fire-mode logic designed for air cooling (stop air handlers, close dampers) may, if applied to liquid cooling, stop CDUs and pumps in unaffected halls, turning a small local fire into a site-wide thermal event for accelerators with seconds of thermal inertia. Cause-and-effect matrices must be re-engineered for liquid-cooled halls (MCF-PHY-022 (Suppression Interfaces)).

Asset value and data sensitivity. Accelerator modules are high-value, portable and attractive to theft; local NVMe in training nodes may hold model weights, checkpoints or training data. Custody of media and decommissioned hardware is therefore both a security and a data-protection matter (MCF-PHY-019 (Media Custody), MCF-PHY-020 (Decommissioning Custody)).

Plant as the soft target. The physical routes to disabling AI capability often run through power and cooling plant — generator yards, fuel fill points, MV substations, dry-cooler compounds — rather than through the data hall. Zoning must protect plant at least as well as the halls it serves (MCF-PHY-009 (Security Zones)).

3 Reference patterns and variants

Pattern (AIDCF-013)	PHY characteristics	PHY questions that dominate
Single-site training facility	Purpose-built halls; slab-on-grade or structural slab; dedicated plant yards	Structural capacity for wet mass; fire zones separating A/B paths
Hyperscale campus	Multiple buildings; shared campus perimeter; central plant	Site hazards affecting whole campus; perimeter and hostile vehicle mitigation; logistics for continuous deployments
Colocation deployment	Tenant suite or cage in provider building	Tenant segregation; disclosure of shared risers and plant; access by provider staff
Modular facility	Prefabricated modules on prepared ground	Anchorage and flood level of modules; inter-module fire separation; module access for replacement
Liquid-cooled high-density cluster	Slab-mounted racks, overhead manifolds and busway	Water zones; rack loading; equipment movement
Brownfield retrofit	Existing building, often raised floor, re-purposed	Raised-floor and slab capacity; fire strategy currency; envelope condition
Edge inference estate	Small rooms in third-party premises	Adjacent risk; access by non-specialist staff; theft of hardware

Security zoning variants are concentric (perimeter → building → secure corridor → critical space → cage/rack), with critical plant assigned to critical-space or higher zones. Fire-protection variants include very early warning aspirating detection with pre-action sprinklers, gaseous agents (inert or chemical) with enclosure integrity, and water mist; lithium-ion rooms add off-gas detection and explosion control (deflagration venting or prevention), guided by large-scale fire test data such as UL 9540A.

4 Required functions and capability linkage

Required Function	Implemented by	Controls
Locate capability where site-wide hazards are within design protection	Site, earthworks, flood defences, perimeter	MCF-PHY-001, MCF-PHY-002, MCF-PHY-003, MCF-PHY-004
Carry, restrain and enclose equipment	Structure, raised floor, anchorage, envelope	MCF-PHY-005, MCF-PHY-006, MCF-PHY-007, MCF-PHY-008
Separate hazards and fault domains spatially	Security, fire, water and tenant zones	MCF-PHY-009, MCF-PHY-010, MCF-PHY-011, MCF-PHY-012
Restrict and observe physical access	Access control, visitor management, video, alarm response	MCF-PHY-013, MCF-PHY-014, MCF-PHY-015, MCF-PHY-016
Maintain custody of assets and data-bearing media	Goods-in, movement, custody store, decommissioning	MCF-PHY-017, MCF-PHY-018, MCF-PHY-019, MCF-PHY-020
Detect and control fire and protect life	Detection, suppression, egress, emergency shutdown	MCF-PHY-021, MCF-PHY-022, MCF-PHY-023, MCF-PHY-024

Capability linkage: the physical layer rarely appears directly in a Service Outcome, but it determines whether redundant power, cooling and network paths are truly independent (fire, water, flood and intrusion as common-mode initiators) and whether recovery is possible (logistics for replacement plant, responder access). Assessors should therefore link PHY controls to the FaultDomain records of POW, THM and NET, and to recovery dependencies in RES.

DG-PHY-REQ-001 The assessor shall overlay the redundant power, cooling and network paths of the declared architecture onto the fire, water and security zone plans, and shall record each zone that hosts both elements of a

redundant pair as a shared FaultDomain or record a U5 UNKNOWN where the overlay cannot be produced.

5 Dependencies, fault domains and common-mode concerns

Physical common-mode initiators are among the most severe because they act on everything in a space at once.

- **Flood and water**: a single flood level, roof failure or burst main can affect both A and B plant if both are at grade or below the same roof (MCF-PHY-001, MCF-PHY-008, MCF-PHY-011).
- **Fire**: shared risers, cable galleries and plant rooms are classic single points where both paths meet (AIDCF-004 X09). Compartmentation is the containment boundary (MCF-PHY-010).
- **External events**: an explosion at an adjacent hazardous installation, a vehicle impact, or a wildfire smoke plume can affect both utility intakes or all heat-rejection plant (MCF-PHY-002).
- **Seismic**: all unanchored equipment of a class responds together (MCF-PHY-007).
- **Physical intrusion**: one breach of a weak plant-zone boundary can reach both generator yards (AIDCF-004 X16; MCF-PHY-009).
- **Emergency shutdown**: an EPO whose actual scope is wider than labelled is a deliberately engineered common-mode switch (MCF-PHY-024).
- **Shared landlord or provider elements**: in colocation, risers, roofs and plant may be shared with other tenants whose activities and incidents propagate (MCF-PHY-004, MCF-PHY-012).

Physical dependencies also include external services: fire and rescue response time, police response for alarm escalation, specialist contractors for heavy lifts, and fuel and water delivery routes.

6 Threats, hazards and failure scenarios with propagation paths

Scenario	Propagation path	Primary safeguards (controls)
Fire disables paired routes (X09)	Cable fire in shared riser → both A and B feeds and fibre routes damaged → loss of hall power and network → outcome loss for weeks	MCF-PHY-010, MCF-PHY-021, MCF-PHY-022
Lithium-ion thermal runaway	Cell fault in rack BBU or UPS string → off-gas not detected → thermal runaway → deflagration → fire spreads to adjacent UPS room → both UPS paths lost; life-safety risk	MCF-PHY-021, MCF-PHY-010, MCF-PHY-023; MCF-POW-010
Fire-mode cooling interlock	Smoke detected in one hall → cause-and-effect stops all CDUs site-wide → accelerators overheat in all halls → mass shutdown	MCF-PHY-022
Inadvertent EPO	Visitor or operator presses unprotected EPO → hall de-energised → uncontrolled shutdown	MCF-PHY-024, MCF-PHY-014
Structural overload	Retrofit liquid cooling raises wet mass → slab deflection and cracking → enforced decant of hall	MCF-PHY-005, MCF-PHY-006
Water ingress (D2 family)	Roof drain blocks in storm → water enters above switchgear → arc fault → path lost	MCF-PHY-008, MCF-PHY-011
Liquid leak with detection gap (X03)	Quick-disconnect fails → coolant reaches busway tap-off → fault → row lost	MCF-PHY-011
Flood exceeds design (X20)	Extreme rainfall above design → generator yard and MV intake flooded → no utility, no generation	MCF-PHY-001; MCF-ENV-023
Unauthorised entry / insider sabotage	Leaver retains access → enters generator yard → disables fuel transfer → generators stop during later outage	MCF-PHY-009, MCF-PHY-013, MCF-PHY-016
Asset removal	Accelerator modules removed during RMA without custody → theft and data exposure	MCF-PHY-017, MCF-PHY-019, MCF-PHY-020
Heavy move incident	Loaded rack tips on ramp → injury; damaged manifolds leak	MCF-PHY-018, MCF-PHY-003

Scenario	Propagation path	Primary safeguards (controls)
Emergency-access conflict	Security locks on containment doors fail secure on power loss → occupants trapped	MCF-PHY-023

7 Safeguards

- **Prevent**: site selection with quantified hazard levels and freeboard; perimeter stand-off and rated hostile vehicle mitigation; structural capacity with utilisation thresholds; anchorage to seismic design level; routing of liquids away from electrical equipment; compartmentation of redundant paths and battery rooms; identity-based access with multi-factor authentication; delivery and custody controls; protected EPO devices.
- **Detect**: aspirating smoke detection designed for high airflow; off-gas detection in battery spaces; leak detection in water zones; intrusion detection and video at zone boundaries; custody reconciliations.
- **Contain**: fire-rated compartments and fire-stopping; drip trays, bunding and drainage; zone boundaries without bypass routes; pre-action sprinklers or gaseous systems with verified enclosure integrity.
- **Correct**: cause-and-effect logic tested end to end; alarm response within declared times; impairment management with compensating measures.
- **Recover**: replacement routes for plant; responder information; decant and relocation plans for halls (with RES).

8 Critical-gate linkage

Gate	Name	CRITICAL controls (this domain)	SUPPORTING controls (this domain)
CG-02	Boundary and architecture integrity	none	MCF-PHY-001, MCF-PHY-005, MCF-PHY-012
CG-03	Critical dependency visibility	none	MCF-PHY-001, MCF-PHY-002, MCF-PHY-004
CG-04	Life safety and hazardous energy	MCF-PHY-021, MCF-PHY-022, MCF-PHY-023, MCF-PHY-024	MCF-PHY-001, MCF-PHY-005, MCF-PHY-006, MCF-PHY-010, MCF-PHY-011, MCF-PHY-018
CG-06	Thermal continuity	none	MCF-PHY-008, MCF-PHY-022
CG-07	Critical control-system integrity	none	MCF-PHY-022, MCF-PHY-024
CG-09	Failure containment	MCF-PHY-010	MCF-PHY-002, MCF-PHY-007, MCF-PHY-009, MCF-PHY-011, MCF-PHY-021, MCF-PHY-024
CG-10	Recovery readiness	none	MCF-PHY-003
CG-13	Responsibility and authority	none	MCF-PHY-012
CG-14	Change control	none	MCF-PHY-006
CG-15	Integrated digital authority over infrastructure	none	MCF-PHY-009, MCF-PHY-013

Notes on the mapping:

- **CG-04** (Life safety and hazardous energy) is carried in PHY by fire detection (MCF-PHY-021), suppression interfaces (MCF-PHY-022), egress and emergency access (MCF-PHY-023) and emergency shutdown governance (MCF-PHY-024). Each is CRITICAL because its failure directly defeats protection of life. The CG-04 evidence profile in AIDCF-008 Part XIII requires E1+E4+E5 at S4, which is why these controls require independent (E5) evidence such as authority-having-jurisdiction acceptance or certified inspection.
- **CG-09** (Failure containment) depends on fire compartmentation separating redundant paths (MCF-PHY-010, CRITICAL); water, security and seismic separation are SUPPORTING.
- **CG-02** (Boundary and architecture integrity) is informed by site hazard assessment, structural capacity and tenant segregation; these are SUPPORTING because CG-02's decisive criteria (boundary approval and as-built reconciliation) are carried by architecture and boundary controls in other domains.
- **CG-15** is informed by physical access control (MCF-PHY-013) and zoning (MCF-PHY-009), because physical access to control rooms, EPMS/BMS servers and consoles is part of authority over infrastructure.

DG-PHY-REQ-002 Where the assessment boundary includes occupied facilities, the assessor shall evaluate CG-04 using every in-scope CRITICAL PHY control and shall not record CG-04 PASS on the basis of a fire certificate or insurer letter alone without the control-level results.

DG-PHY-REQ-003 The assessor shall record the gate-level evaluation required for every SUPPORTING PHY control that is NONCONFORMING before a gate PASS is recorded.

9 Capability-profile parameters

Values marked illustrative show order of magnitude only. Unlabelled values are attributed to the cited source family. Site-specific structural, hazard and fire values must come from the competent engineer of record and the fire strategy.

Control	Parameter	Unit	Illustrative default	Basis
MCF-PHY-001	Design flood annual exceedance probability for critical plant	% per year	0.1 (illustrative)	Capability profile; national flood-risk mapping commonly publishes 1% and 0.1% AEP extents
MCF-PHY-001	Freeboard above design flood level for critical plant	mm	300 (illustrative)	Capability profile; site flood-risk assessment
MCF-PHY-001	Hazard assessment review interval	months	36 (illustrative)	Capability profile
MCF-PHY-002	External exposure assessment radius	km	2 (illustrative)	Capability profile; threat assessment
MCF-PHY-002	Hostile vehicle mitigation rating	vehicle mass / impact speed	7,500 kg at 48 km/h (illustrative)	Threat assessment; rating per IWA 14-1 or PAS 68 test classification
MCF-PHY-002	External exposure review interval	months	12 (illustrative)	Capability profile
MCF-PHY-003	Heaviest declared delivered item on IT route (crated rack)	kg	3,000 (illustrative)	Capability profile; rack OEM shipping data
MCF-PHY-003	Independent access routes for fuel and emergency vehicles	count	2 (illustrative)	Capability profile; emergency services consultation
MCF-PHY-004	Adjacent-risk register review interval	months	12 (illustrative)	Capability profile
MCF-PHY-005	Design uniformly distributed live load, data hall	kN/m²	20 (illustrative)	Structural engineer of record; capability profile for dense liquid-cooled racks

Control	Parameter	Unit	Illustrative default	Basis
MCF-PHY-005	Planned-load utilisation threshold requiring structural re-check	% of rated capacity	80 (illustrative)	Capability profile
MCF-PHY-006	Maximum operating mass per rack position	kg	2,000 (illustrative)	Structural capacity statement; capability profile
MCF-PHY-006	Rack-load reconciliation sample	% of positions	10 (illustrative)	Assessment sampling plan
MCF-PHY-007	Transmitted vibration limit at IT equipment	mm/s RMS	per most restrictive OEM specification	OEM installation requirements
MCF-PHY-008	Envelope inspection interval	months	6 (illustrative)	Capability profile; roofing guidance
MCF-PHY-009	Authentication factors at critical-space zone entry	count	2 (illustrative)	Capability profile; threat assessment
MCF-PHY-010	Fire resistance between compartments holding redundant paths	minutes	120 (illustrative)	Fire strategy; national building regulations; capability profile
MCF-PHY-010	Fire-stopping inspection sample per interval	% of penetrations	20 (illustrative)	Capability profile
MCF-PHY-011	Leak alarm to isolation time (automatic isolation zones)	s	60 (illustrative)	Capability profile; CDU OEM guidance
MCF-PHY-011	Maximum credible leak rate for drainage sizing	L/min	per largest pipe and pump capacity in zone	Thermal design basis
MCF-PHY-013	Critical-zone access review interval	days	90 (illustrative)	Capability profile
MCF-PHY-013	Leaver revocation latency	hours	4 (illustrative)	Capability profile
MCF-PHY-014	Maximum visitors per escort in critical zones	ratio	3:1 (illustrative)	Capability profile
MCF-PHY-015	Recording retention	days	90 (illustrative)	Capability profile; local privacy law limits
MCF-PHY-015	Pixel density for identification views	px/m	250	IEC 62676-4 operational requirement categories
MCF-PHY-016	Alarm acknowledgement time	s	60 (illustrative)	Capability profile
MCF-PHY-016	Attendance time at critical-zone alarm	minutes	5 (illustrative)	Capability profile
MCF-PHY-016	Unannounced response exercises	per year	4 (illustrative)	Capability profile
MCF-PHY-017	Inbound critical shipments with seal inspection	%	100 (illustrative)	Capability profile
MCF-PHY-018	Rack or plant mass requiring an approved move plan	kg	1,500 (illustrative)	Capability profile
MCF-PHY-019	Media custody inventory reconciliation interval	days	30 (illustrative)	Capability profile
MCF-PHY-020	Decommissioned data-bearing devices with per-serial certificate	%	100 (illustrative)	Capability profile

Control	Parameter	Unit	Illustrative default	Basis
MCF-PHY-021	Aspirating detection maximum transport time	s	per design; not greater than the EN 54-20 class limit	EN 54-20 (aspirating smoke detectors); system design
MCF-PHY-021	Detection functional test interval	months	12 (illustrative)	NFPA 72 or national fire code; capability profile
MCF-PHY-021	Maximum detection impairment duration without compensating measure	hours	0 (illustrative)	Capability profile
MCF-PHY-022	Gaseous agent design hold time	minutes	10 (illustrative)	NFPA 2001 or ISO 14520 design practice; fire engineer's design
MCF-PHY-022	Enclosure integrity test interval	months	12 (illustrative)	Capability profile; agent standard
MCF-PHY-023	Emergency lighting full-duration test interval	months	12 (illustrative)	National emergency lighting standard (e.g. EN 50172); capability profile
MCF-PHY-024	Emergency shutdown functional test interval	months	12 (illustrative)	Capability profile; applicable electrical code

DG-PHY-REQ-004 The assessor shall take structural, flood and fire-resistance parameter values from the engineer of record, flood-risk assessment and fire strategy for the site, and shall record any illustrative default retained as an assumption subject to an UNKNOWN Record.

10 Evidence expectations by claim type

Claim type	Assurance Level	Controls	Profile tiers used	Sufficiency floors used
DESIGN	AL2	MCF-PHY-001, MCF-PHY-002, MCF-PHY-004, MCF-PHY-005	CP-1, CP-2	S3
IMPLEMENTATION	AL3	MCF-PHY-006, MCF-PHY-007, MCF-PHY-009, MCF-PHY-010, MCF-PHY-011, MCF-PHY-012, MCF-PHY-015	CP-2, CP-4	S3, S4
OPERATION	AL4	MCF-PHY-003, MCF-PHY-008, MCF-PHY-013, MCF-PHY-014, MCF-PHY-017, MCF-PHY-018, MCF-PHY-019, MCF-PHY-020, MCF-PHY-023	CP-3, CP-5	S3, S4
EFFECTIVENESS	AL5	MCF-PHY-016, MCF-PHY-021, MCF-PHY-022, MCF-PHY-024	CP-4, CP-5	S4

- **DESIGN (AL2)**: hazard assessments, exposure registers, structural statements, adjacency registers. E1 by competent specialists, with E7 analysis (for example overlay of redundant paths on hazard contours). Dates and the source data edition matter: a flood assessment based on superseded mapping is stale (U3).
- **IMPLEMENTATION (AL3)**: as-built zoning, compartmentation, water-zone layouts, anchorage and rack load registers reconciled to the installation by inspection sample. Configuration exports from access-control and video systems (E2).
- **OPERATION (AL4)**: E3 over a representative period: access reviews and leaver reconciliations, visitor records, goods-in and custody logs, inspection records for envelope and fire-stopping, egress inspections, delivery records for the heaviest rack.
- **EFFECTIVENESS (AL5)**: E4 tests: aspirating transport-time tests, end-to-end cause-and-effect tests, enclosure integrity tests, EPO scope tests, unannounced alarm-response exercises. For CG-04 CRITICAL controls, E5 independent evidence is required.

Evidence cautions:

- An insurer survey or a certificate of occupancy is valuable E5 evidence but is evaluated for scope, age and exclusions; it rarely covers liquid-cooling interlocks or rack BBUs.
- Photographs are E3 evidence only when their location, date and integrity are established (AIDCF-008 metadata).
- A fire strategy drawing shows intent (E1); fire-stopping inspection shows implementation (E3); neither shows that detection works (E4).

11 Assessment guidance

1. **Start from outcomes and the fault-domain model.** Identify which redundant paths the PHY controls must keep apart.
2. **Overlay.** Produce the overlay of redundant paths on fire, water and security zones (DG-PHY-REQ-001). This is the most productive single analytical step in PHY.
3. **Walk down.** Inspect plant yards, risers, battery rooms, CDU rooms, roofs (where safe access exists) and a sample of halls. Look for uncontained liquid joints above electrical equipment, unsealed penetrations, zone bypass routes and blocked egress.
4. **Reconcile loads.** Rebuild rack-position loads from the deployment pipeline using operating mass and compare with structural capacity.
5. **Review fire interfaces.** Reconcile the cause-and-effect matrix to fire panel and BMS logic; check what fire mode does to CDUs and pumps.
6. **Test operations.** Reconcile access rights to approvals and leavers; reconcile custody inventories; time alarm responses.
7. **Witness tests.** Witness detection and interface tests where scheduled, from safe positions.
8. **Roll up.** Record results, Findings and UNKNOWN Records; evaluate CG-04 and CG-09 per AIDCF-010.

Sampling guidance: fire-stopping penetrations, rack positions and access holders are large homogeneous populations — sample with weighting to recent works, densest rows and critical zones. Fire detection zones, battery rooms and EPO circuits are assessed exhaustively.

Common pitfalls: accepting a structural statement prepared before liquid cooling; assuming zone continuity without looking above ceilings and below floors; treating plant yards as perimeter-only zones; not checking that rack BBUs are in the battery safety assessment; accepting an EPO test that verified activation but not scope.

12 Testing safety (hazardous-energy testing authorisation)

PHY tests can endanger people (gaseous discharge, isolation of life-safety systems) and capability (EPO, cooling interlocks).

DG-PHY-REQ-005 An assessor shall not operate fire, suppression, EPO, access-control or isolation devices, and shall witness tests only as directed by the person in control of the work.

DG-PHY-REQ-006 A test of suppression interfaces, emergency shutdown or fire-mode cooling interlocks relied on as evidence shall have been performed under a written authorisation by the facility owner's authorised engineer and the fire safety manager, with agent release physically disabled and verified, the alarm receiving centre notified, thermal contingency in place for cooling interlocks, and an approved shutdown window or load transfer for emergency shutdown tests.

Test	Hazard	Precondition
Aspirating transport time (MCF-PHY-021)	False alarm, inadvertent release	Release isolated; ARC notified

Test	Hazard	Precondition
End-to-end cause and effect (MCF-PHY-022)	Agent discharge; loss of cooling	Agent disabled and verified; CDU contingency; method statement
Enclosure integrity (MCF-PHY-022)	Pressure on doors; occupancy	Room cleared and posted
EPO scope (MCF-PHY-024)	Uncontrolled shutdown; electrical hazard	Shutdown window or load transferred; authorised engineer; rollback plan
Door release (MCF-PHY-023)	Security zone exposure	Security staff posted
Alarm response exercise (MCF-PHY-016)	Mis-dispatch of emergency services	Control-room supervisor briefed; no suppression or EPO circuit triggered

Where a life-safety condition is unsafe and unresolved (for example missing detection in a battery room), the CG-04 evidence profile blocks relevant testing as well as assurance until it is resolved.

13 UNKNOWN hot-spots

Control	UNKNOWN rule
MCF-PHY-001 Site Hazard Assessment	If the hazard assessment is absent or predates a material revision of hazard mapping, record a U1 or U3 UNKNOWN against the site and treat flood and seismic protection of critical plant as unverified for CG-02 and CG-04 evaluation.
MCF-PHY-005 Structural Capacity	If the load schedule cannot be reconciled to installed equipment, record a U5 UNKNOWN and treat structural adequacy for the affected area as unverified.
MCF-PHY-010 Fire Zones	If the redundant-path overlay cannot be produced, record a U5 UNKNOWN; CG-09 becomes UNKNOWN for fire as a common-mode initiator.
MCF-PHY-011 Water Zones	If the overlay cannot be produced for a liquid-cooled hall, record a U5 UNKNOWN and carry it into THM leak-management and CG-09 evaluation.
MCF-PHY-012 Tenant Segregation	If the provider will not disclose shared elements, record a U2 UNKNOWN and a U10 dependency UNKNOWN for the affected power and cooling paths.
MCF-PHY-021 Fire Detection	If functional test evidence for detection in a battery room or IT hall is missing or older than the interval, record a US4 UNKNOWN against CG-04; testing that depends on the room (for example load tests) is not authorised until resolved.
MCF-PHY-022 Suppression Interfaces	If the matrix cannot be reconciled to configured logic, or no end-to-end test is current, record a US4 UNKNOWN against CG-04 and refer the cooling interlock gap to CG-06.
MCF-PHY-023 Egress and Emergency Access	If egress in containment or battery rooms cannot be verified, record a US4 UNKNOWN for CG-04.
MCF-PHY-024 Emergency Shutdown Governance	If the actual scope of an EPO function has never been verified by test, record a US4 UNKNOWN against CG-04 and an associated UNKNOWN on CG-09 for unbounded shutdown scope.

Typical causes: landlord or provider refusal to disclose shared risers and plant (U2); fire strategy not updated after liquid-cooling retrofit (U3); no evidence that EPO scope was ever tested (U5); structural load schedule not reconcilable to installed equipment (U5).

DG-PHY-REQ-007 Any UNKNOWN affecting a CG-04 or CG-09 CRITICAL PHY control shall be recorded with severity US4 and shall set the affected gate to UNKNOWN until dispositioned.

14 Interfaces with other domains

Domain	Interface	What PHY needs	What PHY provides
POW	Battery rooms, switchgear, generator yards	Battery safety assessments; arc-flash boundaries; plant layouts	Compartmentation; zoning; structural capacity; responder information
THM	Liquid cooling, CDUs, heat rejection	Pipe routes; fire-mode behaviour of CDUs; leak isolation	Water zones; suppression interfaces; plant-zone protection
ENV	Flood adaptation, refrigerants, waste	Climate-adjusted flood levels; refrigerant hazards	Site hazard data; decommissioning custody
NET	Diverse routes	Fibre route plans	Fire and security separation of diverse routes
SEC	PACS and VMS security	Logical security of access-control and video systems	Physical protection of control rooms and consoles
SUP	Deliveries, provenance	Expected shipments; chain-of-custody requirements	Goods-in records and seal inspections
STO/CMP	Media and hardware	Data classification; sanitisation methods	Custody and decommissioning records
OPS	Work control	Permits, method statements	Move plans; inspection records
RES	Recovery	Relocation and decant plans	Logistics routes for replacement equipment
DEP	Boundaries	Tenant and provider boundary definitions	Shared-element disclosure

15 Informative references

Document-level references; no clause-level reliance is implied.

- EN 50600-1, EN 50600-2-1 (building construction), EN 50600-2-5 (security systems) / ISO/IEC 22237 series
- ANSI/TIA-942 — Telecommunications infrastructure standard for data centres
- NFPA 75 — Fire protection of information technology equipment
- NFPA 76 — Fire protection of telecommunications facilities
- NFPA 72 — National fire alarm and signalling code; EN 54 series — fire detection and fire alarm systems (EN 54-20 aspirating detectors)
- NFPA 2001 — Clean agent fire extinguishing systems; ISO 14520 series — gaseous fire-extinguishing systems
- NFPA 13 — Installation of sprinkler systems
- NFPA 855 — Installation of stationary energy storage systems; UL 9540A — thermal runaway fire propagation test method
- NFPA 101 — Life safety code, or national building regulations; EN 50172 — emergency escape lighting
- FM Global Property Loss Prevention Data Sheets (data centres; energy storage; natural hazards)
- EN 1991 / ASCE/SEI 7 — actions and minimum design loads; EN 1998 — seismic design
- IEC 62305 series — protection against lightning
- IWA 14-1 and PAS 68 — vehicle security barriers
- IEC 60839-11 — electronic access control systems; IEC 62676 series — video surveillance systems; EN 50131 series — intrusion alarm systems
- ISO/IEC 27001 and ISO/IEC 27002 — information security controls (physical controls)
- ISO 28000 — security management systems for the supply chain
- NIST SP 800-88 and IEEE 2883 — media sanitisation

16 Control summary

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-PHY-001	Site Hazard Assessment	DIRECTIVE	AL2	DESIGN	CG-02 SUPPORTING; CG-03 SUPPORTING; CG-04 SUPPORTING
MCF-PHY-002	External Exposure	PREVENTIVE	AL2	DESIGN	CG-09 SUPPORTING; CG-03 SUPPORTING
MCF-PHY-003	Access and Logistics	PREVENTIVE	AL4	OPERATION	CG-10 SUPPORTING
MCF-PHY-004	Adjacent Risk	DIRECTIVE	AL2	DESIGN	CG-03 SUPPORTING
MCF-PHY-005	Structural Capacity	PREVENTIVE	AL2	DESIGN	CG-02 SUPPORTING; CG-04 SUPPORTING
MCF-PHY-006	Rack Loading	PREVENTIVE	AL3	IMPLEMENTATION	CG-04 SUPPORTING; CG-14 SUPPORTING
MCF-PHY-007	Anchorage and Vibration	PREVENTIVE	AL3	IMPLEMENTATION	CG-09 SUPPORTING
MCF-PHY-008	Building Envelope	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING
MCF-PHY-009	Security Zones	PREVENTIVE	AL3	IMPLEMENTATION	CG-15 SUPPORTING; CG-09 SUPPORTING
MCF-PHY-010	Fire Zones	PREVENTIVE	AL3	IMPLEMENTATION	CG-09 CRITICAL; CG-04 SUPPORTING
MCF-PHY-011	Water Zones	PREVENTIVE	AL3	IMPLEMENTATION	CG-09 SUPPORTING; CG-04 SUPPORTING
MCF-PHY-012	Tenant Segregation	PREVENTIVE	AL3	IMPLEMENTATION	CG-13 SUPPORTING; CG-02 SUPPORTING
MCF-PHY-013	Identity-Based Access	PREVENTIVE	AL4	OPERATION	CG-15 SUPPORTING
MCF-PHY-014	Visitor Control	PREVENTIVE	AL4	OPERATION	none
MCF-PHY-015	Surveillance Coverage	DETECTIVE	AL3	IMPLEMENTATION	none
MCF-PHY-016	Alarm Response	DETECTIVE	AL5	EFFECTIVENESS	none
MCF-PHY-017	Delivery Controls	PREVENTIVE	AL4	OPERATION	none
MCF-PHY-018	Equipment Movement	PREVENTIVE	AL4	OPERATION	CG-04 SUPPORTING
MCF-PHY-019	Media Custody	PREVENTIVE	AL4	OPERATION	none
MCF-PHY-020	Decommissioning Custody	PREVENTIVE	AL4	OPERATION	none
MCF-PHY-021	Fire Detection	DETECTIVE	AL5	EFFECTIVENESS	CG-04 CRITICAL; CG-09 SUPPORTING

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-PHY-022	Suppression Interfaces	CORRECTIVE	AL5	EFFECTIVENESS	CG-04 CRITICAL; CG-06 SUPPORTING; CG-07 SUPPORTING
MCF-PHY-023	Egress and Emergency Access	PREVENTIVE	AL4	OPERATION	CG-04 CRITICAL
MCF-PHY-024	Emergency Shutdown Governance	DIRECTIVE	AL5	EFFECTIVENESS	CG-04 CRITICAL; CG-07 SUPPORTING; CG-09 SUPPORTING

17 Worked synthetic example

Synthetic example. The site, figures and results below are invented for illustration and do not describe any real facility or assessment.

Context. Synthetic Site S-2 is a brownfield building retrofitted for a direct-to-chip cluster in Hall 2, with lithium-ion UPS in two UPS rooms and rack BBUs in the cluster racks. The decision question is whether Hall 2 can host the cluster at the declared Service Outcomes through NORMAL, MAINTENANCE and EMERGENCY states. Profile parameters (illustrative for the example): structural re-check threshold 80%; fire resistance between redundant paths 120 minutes; leak alarm to isolation 60 s.

Overlay. The assessor overlays A/B busway, the two UPS rooms and the two fibre routes on the fire-compartment drawings. Both UPS rooms are separate compartments; however both fibre routes and the A and B busway feeders pass through a single riser compartment on level 1.

Selected results.

Control	Claim type	Synthetic observation	Result
MCF-PHY-005	DESIGN	Structural statement predates retrofit and uses dry rack mass; recomputed wet load reaches 92% of capacity in two rows	PARTIAL
MCF-PHY-010	IMPLEMENTATION	Shared level-1 riser hosts both busway feeders and both fibre routes with no rated enclosure	NONCONFORMING
MCF-PHY-011	IMPLEMENTATION	Manifold joints above two tap-off units without drip trays	PARTIAL
MCF-PHY-021	EFFECTIVENESS	Aspirating tests current; rack BBUs not covered by battery safety assessment; no off-gas detection in Hall 2	NONCONFORMING
MCF-PHY-022	EFFECTIVENESS	Cause-and-effect stops all CDUs on any Hall 2 alarm; end-to-end test current	CONFORMING for the matrix as designed; Finding raised against design intent and referred to CG-06
MCF-PHY-024	EFFECTIVENESS	EPO scope never tested since retrofit	UNKNOWN (U5, US4)

Gate Results. CG-04: MCF-PHY-021 NONCONFORMING → FAIL (the US4 UNKNOWN on MCF-PHY-024 would on its own have set CG-04 to UNKNOWN; FAIL takes precedence under the roll-up rule, and the UNKNOWN remains open). MCF-PHY-005 PARTIAL is SUPPORTING to CG-02 and CG-04 and receives a gate-level evaluation: the 92% loading is within capacity, so it does not defeat either gate, but it is carried as a condition. CG-

09: MCF-PHY-010 NONCONFORMING → FAIL. The PHY-022 Finding is evaluated at CG-06 gate level (SUPPORTING): the thermal domain must show whether stopping all CDUs defeats thermal continuity.

Effect on decision. With CG-04 and CG-09 FAIL, the synthetic recommendation is NOT_ASSURED for the Hall 2 outcomes. Actions: enclose one busway feeder and one fibre route in a 120-minute rated enclosure or reroute; include rack BBUs in the battery safety assessment and install the off-gas detection it specifies; obtain a structural re-check; fit drip trays; perform an authorised EPO scope test in the next shutdown window.